Foundation University Islamabad

Department of Software Engineering

Semester Project Proposal

VivaVoice

A RAG-Grounded AI Voice Examiner for Document-Based Oral Assessment

Field	Details
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Course	Software Construction and Development (SCD)
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1. Background and Introduction

Oral examinations (vivas) are one of the most reliable ways to check whether a student truly understands a piece of work, because the student has to explain ideas in their own words instead of selecting an option or copying text. Universities use vivas for semester projects, lab work, internships and final year projects. However, a viva needs an examiner to be present for every student, which makes it slow, tiring and hard to keep consistent across a class of 50 to 60 students.

At the same time, Large Language Models (LLMs) can now read a document, ask questions about it and judge an answer. The problem is that a general AI chatbot is not a fair examiner: it can ask questions from outside the syllabus, it can be convinced by a confident but wrong answer, and it keeps no structured record for the teacher.

VivaVoice is a web-based system in which an examiner uploads a document (a project report, lecture notes or a book chapter) and an AI voice agent conducts an oral exam with each student. The agent speaks the questions, listens to the spoken answers, and evaluates them. Using Retrieval-Augmented Generation (RAG), every question and every grading decision is tied to a specific section of the uploaded document. The agent never asks anything that is not in the document, and the examiner receives a score, feedback and full transcript for each student.

2. Problem Statement

Teachers at Pakistani universities regularly need to assess large classes orally, but they do not have the time to conduct a fair, consistent, one-to-one viva for every student. As a result:

- **Time pressure:** A 10 minute viva for 60 students takes 10 hours of examiner time for one course.
- **Inconsistency:** Students examined later in the day, or by a different examiner, get questions of different difficulty, which makes marking unfair.
- **Weak practice options:** Students have no realistic way to practise a viva on their own report before the real one.
- **Unsafe AI shortcuts:** General-purpose AI chatbots are not grounded in the syllabus, so they drift off topic, are not examiner controlled, and leave no reliable record.

Problem: There is no affordable system that conducts a voice-based oral exam strictly from an examiner-provided document, grades answers by meaning, and gives the examiner transparent, traceable results.

3. Complex Computing Problem (CCP) Justification

According to the HEC/NCEAC criteria, a Complex Computing Problem must show one or more of the characteristics below. VivaVoice clearly satisfies the first two as its primary CCP characteristics and is supported by several others.

CCP Characteristic	How it appears in VivaVoice
Range of Conflicting Requirements (Primary)	Natural conversation vs. strict grounding: the agent must sound like a human examiner yet never leave the document. Fairness vs. anti-cheating: every student must get different questions, but of equal difficulty.

CCP Characteristic	How it appears in VivaVoice
	Strict grading vs. flexibility: correct answers phrased in different words must be accepted while wrong but confident answers must be rejected. Speed vs. accuracy: voice replies must feel instant even though retrieval and LLM calls take time. Usefulness vs. privacy: student voice answers and reports are sensitive data.
Depth of Knowledge (Primary)	Requires knowledge of RAG, text chunking, vector embeddings, similarity search, prompt design for grounding, speech-to-text and text-to-speech, and automated short-answer evaluation. None of these are covered by basic CRUD application knowledge.
Depth of Analysis	Grading a spoken answer is not keyword matching. The system must compare the meaning of the answer with the source passage and decide between correct, partially correct and incorrect, with a reason.
Stakeholder Involvement	Examiner (sets and reviews exams), student (takes or practises exams), administrator (manages users and courses) and the department (needs fair and consistent results). Their needs often differ.
Interdependence	Question quality depends on chunking; grading depends on retrieval; the voice flow depends on speech recognition quality; results depend on all three working together.
Novelty	Combines voice interaction, document-grounded RAG and examiner control in a single oral assessment tool, a combination not offered by the benchmarked systems.
Significant Impact	Saves many hours of examiner time per course and gives every student a fair, repeatable and transparent viva, plus a safe way to practise.

4. Business Objectives

1. Reduce examiner time for oral assessments by letting the AI agent conduct the first-round viva for every student.
2. Ensure 100% of questions are generated from, and traceable to, the uploaded document.
3. Give every student a consistent difficulty level, regardless of when they take the exam.
4. Provide examiners with a transparent report: score, feedback, transcript and source section for every question.
5. Provide students with a practice mode to rehearse a viva on their own report before the real exam.

5. Preliminary Requirements (SRS Summary)

5.1 Functional Requirements

ID	Requirement
FR-01	The system shall allow users to register and log in with one of three roles: Examiner, Student or Admin.
FR-02	The system shall allow an examiner to upload a PDF or DOCX document of up to 20 MB as the exam source.
FR-03	The system shall extract the text of the uploaded document, split it into chunks and store an

ID	Requirement
	embedding vector for each chunk.
FR-04	The system shall allow an examiner to set the number of questions (3 to 15), difficulty (Easy, Medium, Hard) and time per answer (30 to 180 seconds).
FR-05	The system shall generate a unique exam code that students use to join the exam.
FR-06	The system shall speak each question aloud to the student using text-to-speech.
FR-07	The system shall convert the student's spoken answer to text and display the live transcript on screen.
FR-08	The system shall generate every question only from chunks retrieved from the examiner's document.
FR-09	The system shall refuse any student request for help or off-topic discussion and repeat or move to the next question.
FR-10	The system shall grade each answer against its source chunk as Correct, Partially Correct or Incorrect, with a score from 0 to 10 and a one-line reason.
FR-11	The system shall give each student a different set of questions drawn from different chunks of the same document.
FR-12	The system shall record a tab switch or window change during the exam and flag it in the result.
FR-13	The system shall show the examiner a results dashboard with each student's total score, per-question feedback, transcript and source section.
FR-14	The system shall allow the examiner to override any AI score before results are published.
FR-15	The system shall provide a Practice Mode in which a student uploads their own document and receives a mock viva with feedback.
FR-16	The system shall provide a typing fallback when the microphone is unavailable or speech recognition fails.

5.2 Non-Functional Requirements

ID	Attribute	Requirement
NFR-01	Performance	The agent shall speak the next question within 5 seconds of the student finishing an answer.
NFR-02	Accuracy	At least 90% of AI grades shall match the examiner's grade on a test set of 50 answers.
NFR-03	Security	Passwords shall be stored as bcrypt hashes and all API routes shall require a valid JWT token.
NFR-04	Privacy	Uploaded documents and transcripts shall be visible only to the owning examiner and the relevant student.
NFR-05	Usability	A first-time student shall be able to start an exam within 2 minutes using only the exam code.
NFR-06	Compatibility	The exam screen shall work on the latest versions of Google Chrome and Microsoft Edge on desktop.
NFR-07	Cost	The system shall run entirely on free tiers of the chosen services for a class of up to 60 students.

6. Current Situation and Opportunity

6.1 Current Situation

- Vivas are conducted manually, one student at a time, with no written record of what was asked or why marks were given.
- Written quizzes (paper, Google Forms, LMS) test recall but not the ability to explain ideas verbally.
- Students use general AI chatbots for practice, which are not limited to their syllabus and do not simulate exam conditions.

6.2 Opportunity

Free LLM and embedding APIs, free cloud databases with vector search, and speech recognition built into modern browsers now make it possible to build a voice examiner at almost zero cost. VivaVoice uses these to give teachers a fair, consistent and traceable first-round viva, and gives students a realistic way to practise.

7. Benchmarking with Existing Systems

VivaVoice was compared with four existing systems that students and teachers currently use for AI-assisted learning or assessment.

Feature	NotebookLM	ChatGPT Voice	HireVue	Moodle AI Quiz Plugins	VivaVoice
Voice-based oral exam	Partial	Yes	Yes	No	Yes
Questions only from uploaded document	Yes	No	No	Yes	Yes
Examiner sets count, difficulty, time	No	No	Yes	Yes	Yes
Automatic grading with feedback	Partial	Partial	Yes	Partial	Yes
Source section shown per question	Yes	No	No	No	Yes
Different questions per student	No	No	Partial	Yes	Yes
Examiner can override AI score	No	No	Yes	Yes	Yes
Student practice mode	Yes	Yes	No	No	Yes
Built for academic oral exams	No	No	No	No	Yes
Free for a university class	Yes	Partial	No	Yes	Yes

Table 7.1: Feature comparison (based on publicly available feature information, October 2026). Partial = available in a limited or indirect form.

Key gap: No existing system combines a spoken exam, strict grounding in an examiner-uploaded document, and examiner-controlled, traceable grading. NotebookLM is grounded but is a study tool,

not an exam. ChatGPT Voice can talk but is not limited to the document. HireVue is built for job interviews, not academic content. Moodle plugins generate written questions only.

Literature basis: The RAG approach used in VivaVoice follows Lewis et al. (2020), who showed that retrieving relevant passages before generation makes model outputs more factual and traceable to their sources.

8. Proposed System

8.1 Main Features

- **Examiner Portal:** upload document, configure exam, share exam code, review and override results.
- **Document Ingestion and RAG Pipeline:** text extraction, chunking, embeddings and vector storage.
- **AI Voice Exam Agent:** speaks questions, listens to answers, follows the exam flow and keeps the conversation inside the document.
- **Grounding Guard:** checks that every generated question is supported by its source chunk and blocks off-topic or help-seeking requests.
- **Evaluation Engine:** scores each answer by meaning, with a reason and the source passage.
- **Results Dashboard:** per-student and class-level results, transcripts and integrity flags.
- **Practice Mode:** students rehearse a viva on their own report.

8.2 System Architecture

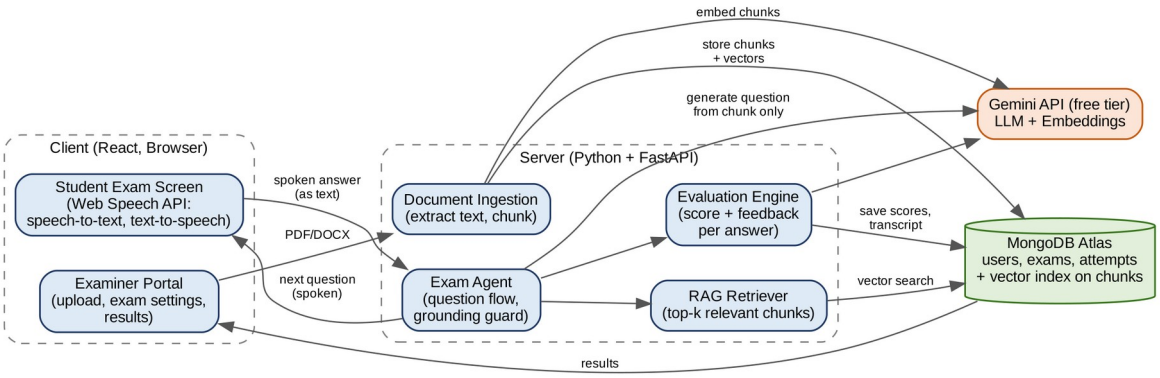


Figure 8.1: VivaVoice high-level architecture

8.3 Exam Flow

1. The examiner uploads a document and sets the exam settings.
2. The server extracts the text, splits it into chunks, creates embeddings and stores them in MongoDB Atlas.
3. A student joins with the exam code and grants microphone access.
4. The exam agent picks an unused chunk, generates one question strictly from it, and speaks it.
5. The student answers by voice; the browser converts speech to text.
6. The evaluation engine compares the answer with the source chunk and stores the score, reason and transcript.

7. Steps 4 to 6 repeat until the set number of questions is reached; the examiner then reviews the results.

9. Tools and Technologies

Layer	Technology
Frontend	React.js, Tailwind CSS
Voice	Web Speech API (built into Chrome/Edge): speech-to-text and text-to-speech, no API key
Backend	Python with FastAPI (REST API)
AI Model	Google Gemini API, Flash model (free tier) for question generation and grading
Embeddings	Gemini embedding model (free tier)
Database	MongoDB Atlas (free cluster) with Atlas Vector Search for chunk retrieval
Document Parsing	PyMuPDF (PDF), python-docx (DOCX)
Authentication	JWT and bcrypt
Version Control	Git and GitHub
Deployment	Vercel (frontend) and Render (backend), free tiers
Design and Testing	Figma or draw.io (UI and diagrams), Postman (API testing), pytest (unit tests)

10. Options and Recommendation

Option	Description	Limitation
A. Manual viva	Examiner questions each student in person.	Slow, inconsistent, no record.
B. Written AI quiz	AI generates MCQs or short questions from notes.	Does not test verbal explanation.
C. General AI chatbot	Students practise with a chatbot by voice.	Not grounded, not examiner controlled.
D. VivaVoice (Recommended)	**Grounded AI voice examiner with examiner control and traceable grading.**	**Depends on speech recognition quality (mitigated by typing fallback).**

Recommendation: Option D is recommended because it is the only option that is fair, consistent, voice-based and traceable, and it can be built and run at zero cost.

11. Schedule Estimate

Week	Activity
Week 1 to 2	Requirement gathering, SRS and Business Case finalisation
Week 3 to 4	System design: use cases, class and sequence diagrams, database schema, UI wireframes

Week	Activity
Week 5 to 6	Authentication, roles, examiner portal and document upload
Week 7 to 8	RAG pipeline: text extraction, chunking, embeddings and vector search
Week 9 to 10	Voice exam agent: question generation, speech-to-text, text-to-speech, grounding guard
Week 11	Evaluation engine and examiner override
Week 12	Results dashboard and practice mode
Week 13	Testing (unit, integration, grading accuracy check) and bug fixing
Week 14	Deployment, final report and project demonstration

12. Potential Risks and Mitigation Plan

Risk	Impact	Mitigation
AI asks a question not in the document (hallucination)	High	Generate from one retrieved chunk only; run a second check that the question is answerable from that chunk; store the chunk with the question.
Speech recognition mishears accents or mixed Urdu-English	Medium	Show live transcript, allow the student to re-record once, and provide a typing fallback.
Unfair AI grading	High	Grade against the source chunk with a fixed rubric; examiner can override every score before publishing.
Student reads answers from the document	Medium	Time limit per answer, tab-switch detection and different questions per student.
Free API rate limits exceeded	Medium	Cache embeddings, pre-generate a question pool per exam, and space requests; switch to another free LLM provider if needed.
Privacy of student data	Medium	Role-based access, JWT authentication, and deletion of uploaded documents after the exam if the examiner chooses.

13. Expected Outcome

By the end of the semester, VivaVoice will be a working web application in which an examiner uploads a document and receives fair, traceable viva results for the whole class, and in which students can practise a viva on their own work. The project demonstrates a Complex Computing Problem through its conflicting requirements and the depth of knowledge it requires, while keeping the implementation within the scope of a single developer using free tools.

References

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